🔷 1. Introduction

Set theory operations in relational algebra are **binary operations** (work on two relations).

They are used to:

- Combine relations
- Compare relations
- Find common or different tuples

These operations are based on **mathematical set theory**.

🔷 2. Union Compatibility (Type Compatibility)

Before applying any set operation:

Two relations must be **Union Compatible**.

✅ Conditions:

If A(P1, P2, …, Pn)
and B(Q1, Q2, …, Qn)

Then:

1. Both must have **same number of attributes (degree)**
2. Corresponding attributes must have **same domain**

👉 domain(Pi) = domain(Qi)

🔷 3. UNION Operation (∪)

📌 Notation:

A ∪ B

📌 Definition:

Returns all tuples present in:

- A
- B
- Or both

Duplicates are removed.

📌 Example:

R = {(1, 'Alice'), (2, 'Bob')}
S = {(2, 'Bob'), (3, 'Charlie')}

R ∪ S = {(1, 'Alice'), (2, 'Bob'), (3, 'Charlie')}

📌 **Properties:**

- ✅ Commutative
  A ∪ B = B ∪ A
- ✅ Associative
  A ∪ (B ∪ C) = (A ∪ B) ∪ C

📌 **SQL Equivalent:**

- UNION → Removes duplicates
- UNION ALL → Keeps duplicates

🔷 **4. INTERSECTION Operation (∩)**

📌 **Notation:**

A ∩ B

📌 **Definition:**

Returns only tuples present in **both** A and B.

📌 **Example:**

R ∩ S = {(2, 'Bob')}

📌 **Properties:**

- ✅ Commutative
  A ∩ B = B ∩ A
- ✅ Associative
  A ∩ (B ∩ C) = (A ∩ B) ∩ C

📌 **Can Be Derived As:**

A ∩ B = ((A ∪ B) − (A − B)) − (B − A)

📌 **SQL Equivalent:**

- INTERSECT
- INTERSECT ALL

🔷 **5. MINUS / SET DIFFERENCE (−)**

📌 **Notation:**

A − B

📌 **Definition:**

Returns tuples present in **A but not in B**

📌 **Example:**

R − S = {(1, 'Alice')}

📌 **Important Property:**

❌ Not Commutative
A − B ≠ B − A

📌 **SQL Equivalent:**

- EXCEPT
- EXCEPT ALL

🔷 **Example with Student & Faculty Tables**

Student Table

First	Last
Aisha	Arora
Bikash	Dutta
Makku	Singh
Raju	Chopra

Faculty Table

FirstN	LastN
Raj	Kumar
Honey	Chand
Makku	Singh
Karan	Rao

✅ 1. Student ∪ Faculty

All unique names from both tables:

Aisha Arora
Bikash Dutta
Makku Singh
Raju Chopra
Raj Kumar
Honey Chand
Karan Rao

✅ 2. Student ∩ Faculty

Common entries:

Makku Singh

✅ 3. Student – Faculty

Students not in Faculty:

Aisha Arora
Bikash Dutta
Raju Chopra

🔥 Summary Table

Operation	Symbol	Meaning	Commutative?	SQL Equivalent
Union	∪	All tuples	✅ Yes	UNION
Intersection	∩	Common tuples	✅ Yes	INTERSECT
Minus	–	A but not B	❌ No	EXCEPT

🎯 Key Points for Exams

✓ Set operations require Union Compatibility
✓ UNION removes duplicates
✓ INTERSECTION gives common tuples
✓ MINUS gives difference
✓ Only MINUS is NOT commutative
✓ SQL equivalents: UNION, INTERSECT, EXCEPT